A REDUCED ONE-DIMENSIONAL NAGATA LOCAL RING HAS VANISHING DELTA-INVARIANT EXACTLY WHEN IT IS A DISCRETE VALUATION RING

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Allowed standard prerequisites (editorial).

- Reducedness and normality via integral closedness in the total ring of fractions. Stacks Project, Algebra, Lemma algebra-lemma-characterize-reduced-ring-normal (tag 030W). Turns vanishing of the delta-invariant into normality of A.
- Characterization of discrete valuation rings. Stacks Project, Algebra, Lemma algebra-lemma-characterize-dvr (tag 034Z). Concludes that a one-dimensional Noetherian normal local ring is a discrete valuation ring.

Source reference. Original reference 030C: sources/stacks-chow/tags.txt, source lines 3943–3943. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00PD: sources/stacks-chow/tags.txt, source lines 927–927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A be a reduced Nagata local ring of dimension* 1*. The δ-invariant of A is* 0 *if and only if A is a discrete valuation ring.*

Proof. If A is a discrete valuation ring, then A is normal and the ring A' is equal to A. Conversely, if the δ-invariant of A is 0, then A is integrally closed in its total ring of fractions which implies that A is normal (Algebra, Lemma 030C) and this forces A to be a discrete valuation ring by Algebra, Lemma 00PD. □

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